SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Student Performance Prediction System

Machine Learning Regression Pipeline for Academic Outcome Forecasting

DOCUMENT VERSION	MODEL TYPE	TARGET VARIABLE
v1.0 (October 2026)	Polynomial Linear Regression	Exam_Score

PROJECT SCOPE

Executive Overview

End-to-end ML workflow designed to analyze student indicators and predict test scores.

Primary Purpose

Automate student performance analysis using historical academic and demographic predictors to estimate final exam outcomes.

Implementation

Delivered as an interactive Python Jupyter Notebook utilizing standard Scikit-Learn pipelines and tabular data structures.

Core Mechanism

Processes feature interactions with degree-2 polynomials and one-hot encoding prior to linear estimator execution.

SYSTEM ARCHITECTURE

End-to-End Processing Workflow

Sequential data processing steps from raw dataset ingestion to metrics visualizer.

1

Data Ingestion

Loads CSV dataset and splits features from target Exam_Score.

2

Preprocessing

Imputes missing values and scales numerical/categorical factors.

3

Feature Expansion

Generates degree-2 polynomial features for nonlinear signals.

4

Model Fitting

Trains Linear Regression estimator on 80% split data.

DATA ENGINEERING

Feature Transformation Rules

Differentiated transformation pipelines for numerical vs categorical variables.

Numerical Pipeline

Missing Values: Median imputation strategy

Scaling: StandardScaler standardizes variance

Polynomials: Degree-2 interaction term expansion

Bias Handling: `include_bias=False` parameter

Categorical Pipeline

Missing Values: Most frequent value imputation

Encoding: OneHotEncoder converts categories

Unknown Values: `handle_unknown='ignore'` set

Format: Sparse binary column generation

REQUIREMENTS SPECIFICATION

Key Functional Requirements

Core capabilities mandated for the machine learning notebook execution.

FR-01 to FR-03: Ingestion & Target

Loads `StudentPerformanceFactors.csv`, isolates `Exam_Score` target, displays dataset dimensions.

FR-05 to FR-07: Data Preprocessing

Imputes missing cells, scales features, generates degree-2 polynomials, and encodes categories safely.

FR-08 & FR-09: Split & Training

Splits dataset 80/20 with fixed `random_state=42` and fits standard Linear Regression model.

FR-11 & FR-12: Evaluation & Plot

Reports R^2, MAE, RMSE metrics and renders actual vs predicted scatter plot with reference line.

PERFORMANCE VERIFICATION

Model Evaluation Metrics

Standardized metrics collected during execution to quantify model precision.

R^2

Coefficient of Determination

Measures variance explained by predictions on test split.

MAE

Mean Absolute Error

Quantifies average magnitude of absolute target errors.

RMSE

Root Mean Squared Error

Penalizes larger prediction gaps via squared residual root.

QUALITY ATTRIBUTES

Non-Functional Constraints

Operational boundaries and system limitations documented in SRS.

Reproducibility

Fixed seed `random_state=42` guarantees identical train/test splits across runs.

Robustness

OneHotEncoder ignores unseen categorical classes without breaking execution.

Scope Limits

Offline experiment artifact; lacks web API, persistent storage, or online serving.

NEXT STEPS

Future Enhancements & Roadmap

Recommended upgrades to transition notebook prototype into production service.

Model Baseline Comparison

Evaluate against Ridge, Lasso, and Tree-based Ensembles via Cross-Validation.

Model Persistence

Serialize trained ColumnTransformer and Linear Model using standard Joblib tools.

Web Interface API

Build FastAPI/Flask application for real-time student performance queries.

Validation & Safeguards

Implement strict schema verification for inputs and out-of-bounds score alerts.